Lightweight Feature Matching via Knowledge Distillation from LoFTR

ECE1508 Applied Deep Learning — Winter 2026

Jerry Chen Spiro Li Weijie Zhu

The Edward S. Rogers Sr. Department of Electrical & Computer Engineering
University of Toronto

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Motivation

Feature matching is the foundation of visual localization, SLAM, and 3D reconstruction.

LoFTR (Sun et al., 2021)

- State-of-the-art detector-free matcher
- Transformer-based, $\sim$**12M** parameters
- Too heavy for mobile / edge deployment

Our Goal

- Compress LoFTR via **knowledge distillation**
- Student models with $<$1M parameters
- $\sim$**20**$\times$ parameter reduction

Can a tiny student learn LoFTR's matching ability?

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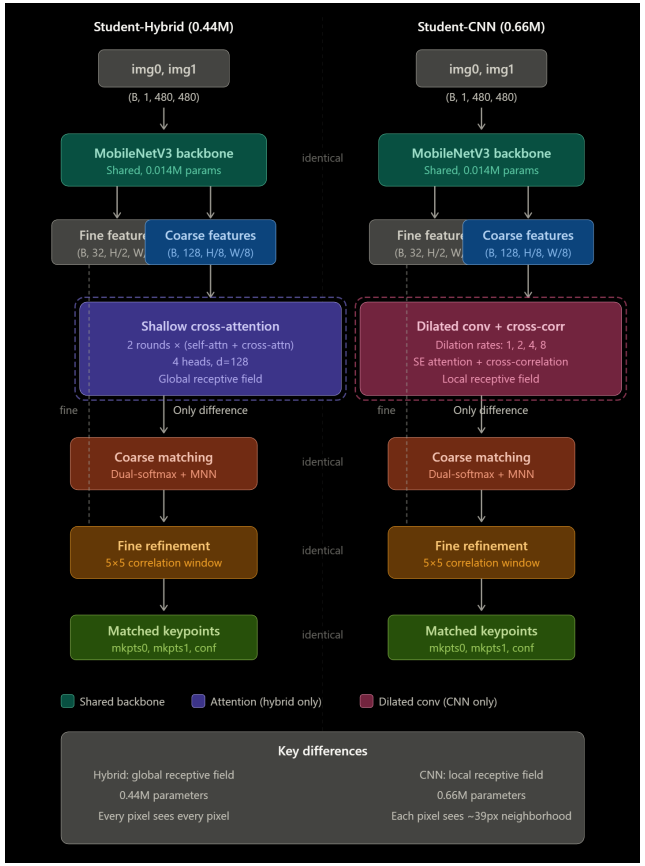
Student Architecture



Student-Hybrid (0.44M)
Shallow cross-attention
Global receptive field

Student-CNN (0.66M)
Dilated convolutions
Local receptive field

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Method & Results

$\mathcal{L}_{total} = 1.0 \cdot \underbrace{\mathcal{L}^{KD}_{coarse}}_{\text{KL div}} + 0.5 \cdot \underbrace{\mathcal{L}^{KD}_{feat}}_{\text{MSE}} + 1.0 \cdot \underbrace{\mathcal{L}_{GT}}_{\text{Focal}}$ **Setup:** HPatches, 1k pairs/epoch, AdamW 3e-4, 20 ep

Hybrid 0.44M CNN 0.66M

	Threshold = 0.2			**Threshold = 0.02**		
Model	@3	@5	@10	@3	@5	@10
Hybrid (KD+GT)	0.00	0.04	0.33	12.65	24.03	**36.17**
CNN (KD+GT)	1.58	3.91	7.80	12.59	22.59	34.34
Hybrid (GT-only)	9.95	18.36	31.38	9.95	18.36	31.38
CNN (GT-only)	**13.18**	**22.70**	**34.34**	13.18	22.70	34.34
LoFTR Teacher	51.82	66.62	77.99	—	—	—

1. **Confidence calibration:** KD flattens confidence ⇒ threshold 0.2 fails; at 0.02 Hybrid recovers to **36.17%**
2. **CNN > Attention:** CNN wins in most configs (more params, shallow attn too weak, local features suffice)
3. **Gap:** best student 36% vs teacher 78% (20× compression + synthetic data domain gap)

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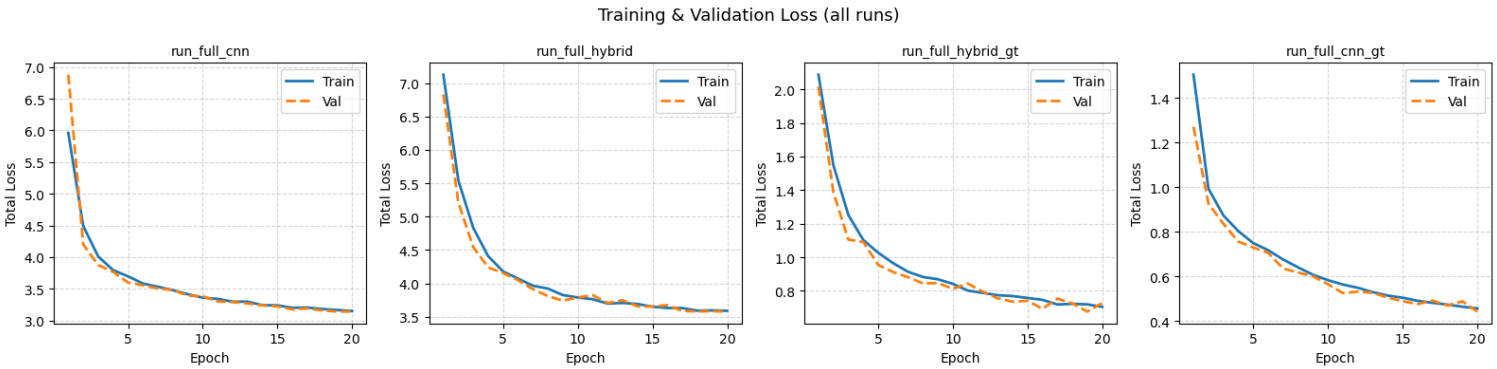
Conclusions & Future Work

Conclusions

1. KD **does transfer** matching knowledge, but requires **confidence recalibration**
2. Convolutional student **outperforms** shallow attention at this scale
3. Training data quality is the **primary bottleneck**, not architecture

Future Work: MegaDepth real 3D data — deeper attention — learnable **confidence calibration layer**



Training & validation loss — KD+GT (≈3.1) vs GT-only (≈0.5)

Thank you! Questions?

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